

Zhixuan Feng

Department of Economics, Poole College of Management
North Carolina State University, Raleigh, NC 27695, US

Email: zfeng9@ncsu.edu

Phone: +1 (984) 867-9772

Location: Raleigh, NC

EDUCATION

Ph.D. in Economics , North Carolina State University, Raleigh, NC	2022 – 2027 (Expected)
Specialization: Financial Econometrics, Time Series Analysis	
Master of Statistics , North Carolina State University, Raleigh, NC	2025 – 2026
Master of Economics , North Carolina State University, Raleigh, NC	2021 – 2023
B.A. in Finance & B.S. in Electronic Information Engineering , UESTC, Chengdu, China	2017 – 2021

RESEARCH FIELDS

Financial Econometrics, High-Frequency Volatility Modeling, Time Series, Machine Learning

JOB MARKET PAPER

“Regime-Switching in Multivariate Realized GARCH: A Block-Constrained Hidden Markov Approach”

- Abstract:** This paper proposes a two-stage Hidden Markov Model (HMM) with a block-structure specification to capture structural shifts in the joint dynamics of returns and realized covariance matrices. By leveraging the Hamilton Filter and the Louis Formula for recursive state inference and analytical parameter estimation within a Multivariate Realized GARCH framework, the model alleviates dimensionality constraints while substantially improving out-of-sample density forecasts and Value-at-Risk predictions during market stress regimes.

WORKING PAPERS

“Forecasting High-Dimensional Realized Covariance: Regularization and Non-synchronous Trading”

- Constructs daily positive semi-definite realized covariance ($RCOV$) matrices via refreshing-time synchronization and multivariate realized kernels to correct for microstructure noise and asynchronous trading effects.
- Evaluates forecasting performance across Vector Autoregressions (VAR), Heterogeneous Autoregressive (HAR) models, and regularized machine learning techniques (Lasso, Elastic Net), establishing that shrinkage methods effectively suppress parameter estimation error in large-scale cross-sectional dispersion.

INDUSTRY EXPERIENCE

Quantitative Research Intern , Southwest Securities Co., Ltd., Chongqing, China	07/2026 – 08/2026
- Conducted quantitative research within the Equity Investment Department, extracting alpha signals from high-frequency futures data combined with underlying equity dynamics. - Engineered volatility and momentum metrics to capture cross-asset mispricing; executed comprehensive backtesting across multiple rolling estimation windows. - Evaluated factor effectiveness via Information Coefficient (IC/IR), turnover ratios, and risk-adjusted returns, identifying robust and decay-resistant alpha factors for live allocation strategies.	
FinTech Analyst Intern , China Construction Bank, Chongqing, China	07/2019 – 09/2019
- Processed large-scale ETC customer transaction datasets using big data frameworks to optimize targeting strategies for personalized marketing campaigns. - Assisted municipal government liaisons and FinTech sector leads in technical integration and documentation during the Smart China Expo Summit Forum.	

TEACHING EXPERIENCE

Independent Instructor, North Carolina State University 2024 Fall

- Principles of Macroeconomics (EC 202): Designed curriculum, delivered lectures, and evaluated undergraduate students on macroeconomic modeling, monetary/fiscal policy, and aggregate market dynamics.

Graduate Teaching Assistant, North Carolina State University 2022 – Present

- Undergraduate Economics Courses: Facilitated weekly recitation sessions, organized exam review lectures, handled course grading, and held office hours.

TECHNICAL SKILLS & INFORMATION

Programming	: MATLAB, Python (NumPy, SciPy, Pandas, Scikit-Learn), R, SAS, Stata
Methodologies	: Time Series, Hidden Markov Models, High-Frequency Econometrics, Volatility Modeling, Factor Alpha Mining, Machine Learning
Tools & Systems	: LaTeX, Git
Languages	: English (Fluent), Chinese (Native)